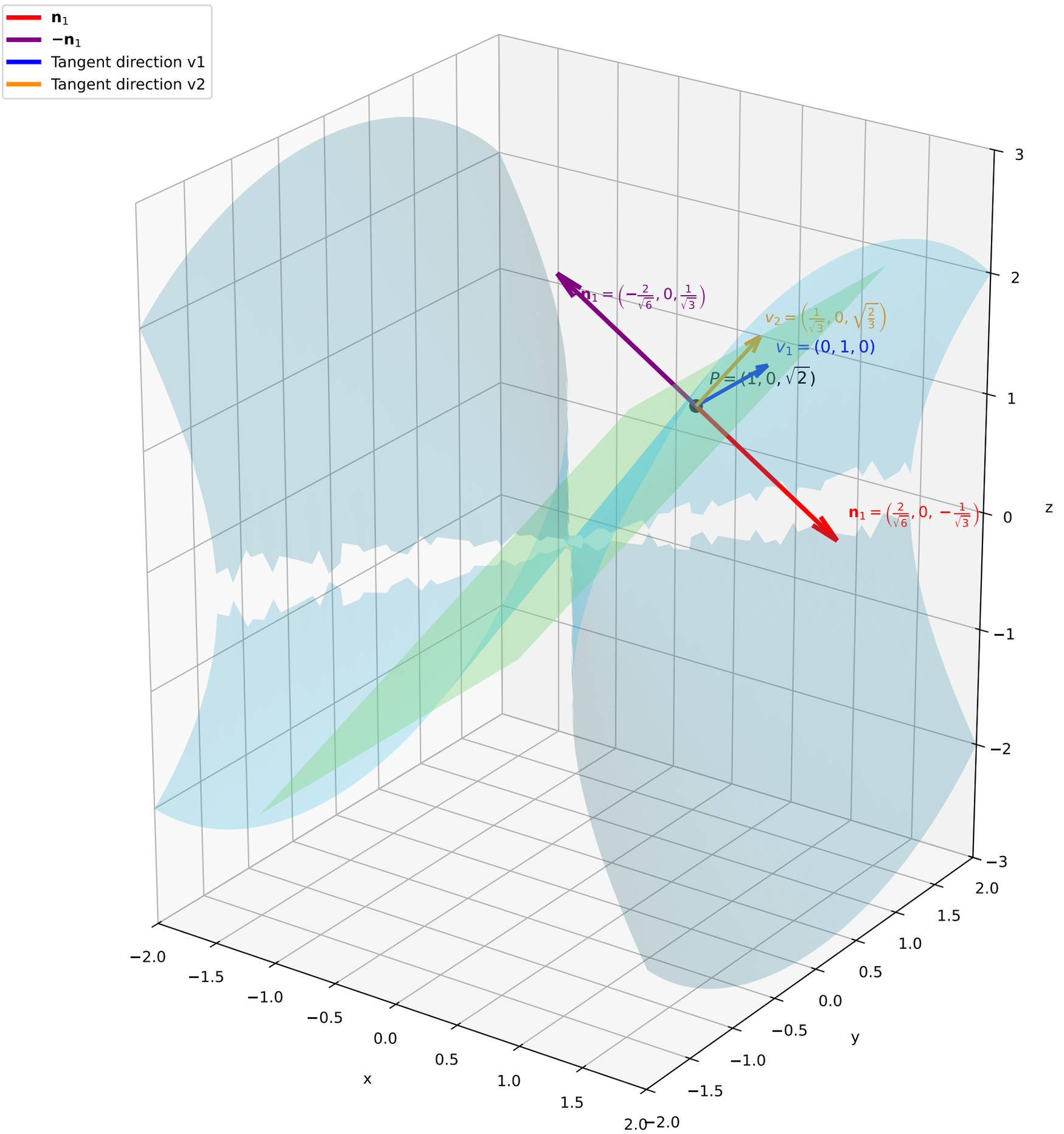

Quadratic Surface: Tangent Plane and Normals
$\mathbf{n}_1$
$-\mathbf{n}_1$
Tangent direction v1
Tangent direction v2
$-\mathbf{n}_1 = \left(-\frac{2}{\sqrt{6}}, 0, \frac{1}{\sqrt{3}}\right)$
$v_2 = \left(\frac{1}{\sqrt{3}}, 0, \sqrt{\frac{2}{3}}\right)$
$v_1 = (0, 1, 0)$
$P = (1, 0, \sqrt{2})$
$\mathbf{n}_1 = \left(\frac{2}{\sqrt{6}}, 0, -\frac{1}{\sqrt{3}}\right)$
x
y
z
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
3
2
1
0
−1
−2
−3